VibeScent

Multimodal Fragrance Recommender

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System Architecture

01	02	03	04	05	06
DATA PREP	**INPUT LAYER**	**IMAGE + TEXT PROCESSESING**	**VIBE VECTOR EXTRACTION**	**RECOMMEND**	**WEB SCRAPE**
Collecting + Processing Data	Image + Text Input	Vision + Sentence Transformers	5-Dimensonal Classification	Weighted Sum + Rank	SerpAPI + Google Shop

TECH STACK

- llama-3.1-8b-instant
- all-MiniLM-L6-v2
- gemma-3-27b-it
- CLIP ViT-L/14
- Cosine Similarity
- SerpAPI requests

Stage 01 — Data Prep

LLM VIBE EXTRACTION | llama-3.1-8b-instant | all-MiniLM-L6-v2

SOURCE DATASETS

- **Parfumo** — *TidyTuesday*
- **Fragrantica Clean** — *Kaggle / olgagmiufana1*
- **Rawanalqarni** — *Kaggle*
- **rdemarqui** — *Kaggle*
- **Ayush** — *Kaggle*

~36,000
FRAGRANCES

01 Text Embedding

all-MiniLM-L6-v2

- Combines fragrance notes + accords
- Embeds each fragrance scent profile
- Same model as inference pipeline

384-d
EMBEDDING DIM

02 LLM Vibe Tag

llama-3.1-8b-instant

- Vibe extracted from each fragrance
- 5 new vibe columns appended
- Allows for later comparison with user query

36K
DESCRIPTIONS GENERATED

OUTPUT SCHEMA — vibescent_enriched.csv

Formality	Season	Gender	Time_of_Day	Frequency
0.0 – 1.0	Spring · Summer · Autumn · Winter	Male · Female · Unisex	Day · Night	Occasionally · Everyday

Stage 02 — Input Layer

MULTI-MODAL INPUT

MODALITY 01

Outfit Image

› **Input:** The user uploads an outfit they plan to wear

› **Goal:** Identify the outfit and fragrances that match

› **Pipeline:** Feeds into a vision transformer as raw pixel tensor

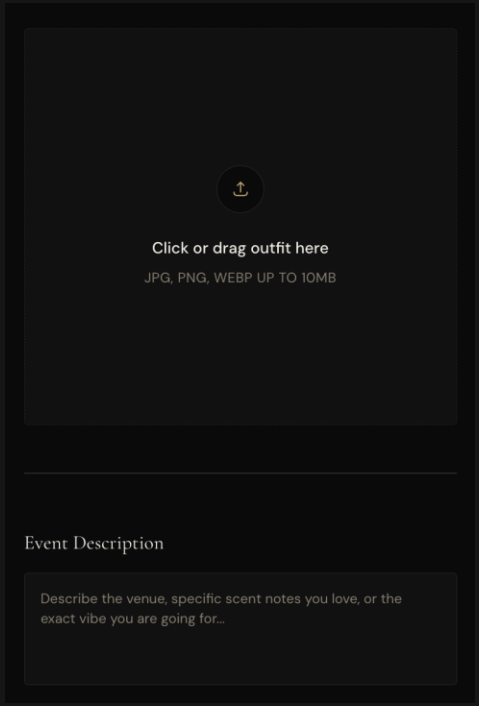
MODALITY 02

Occasion Description

› **Input:** The user describes the occasion they are dressing for

› **Goal:** Understand the occasion and choose fragrances that are appropriate

› **Pipeline:** Feeds into a sentence transformer as a string



Stage 03 — Image + Text Processing

CONTEXT | openai/clip-vit-large-patch14 | all-MiniLM-L6-v2

VISION TRANSFORMER

CLIP ViT-L / 14

Vision Backbone:	Vision Transformer (ViT-L/14)
Patch Size:	14 × 14 pixels
Image Resolution:	224 × 224 (CLIPProcessor)
Embedding Dim:	768-d L2-normalized vector
Text Encoder:	Transformer (77 token context)
Contrastive Loss:	Softmax cross-entropy (ITAL)

SENTENCE TRANSFORMER

all-MiniLM-L6-v2

Model:	all-MiniLM-L6-v2
Architecture:	BERT-based (6-layer MiniLM)
Output Dim:	384-d vector
Max Tokens:	256 (WordPiece tokenizer)
Pooling:	Mean pooling over embeddings
Training Obj.:	Multiple Negatives Ranking Loss

Models loaded once at FastAPI startup to eliminate cold-start latency on user query

Stage 04 — Vibe Vector Extraction

5-DIM VIBE REPRESENTATION

FORMALITY

Formality

Scale: casual → smart casual → formal

Type: Float

SEASON

Season

Classes: spring · summer · fall · winter

Type: Categorical

GENDER

Gender

Classes: male · female · neutral

Type: Categorical

TIME OF DAY

Time of Day

Classes: day · night

Type: Binary

FREQUENCY

Frequency

Classes: occasional · everyday

Type: Binary

Stage 05 — Recommend

MULTIMODAL LATE FUSION

gemma-3-27b-it

Text

- Qwen3 embeds context input
- Text embeddings vs corpus
- Cosine similarity outputs a scalar 0-1

27.5%
TEXT WEIGHTING

Image

- CLIP ViT-L / 14 embeds image input
- Image embeddings vs corpus
- Cosine similarity outputs a scalar 0-1

45%
IMAGE WEIGHTING

Structured

- Context mapped to target feature values
- Target values vs structured CSV columns
- Absolute distance avg outputs a scalar 0-1

27.5%
STRUCTURED WEIGHTING

Fused Score = 0.275 (Text) + 0.45 (Image) + 0.275 (Structured)

Reranking

- gemma-3-27b-it takes the top 10 fragrances and runs a mini-pipeline for each candidate
- Compares the input image and query with each fragrance candidate and outputs the top 3 final recommendations

Stage 06 — Web Scrape

SERPAPI SCRAPER

01

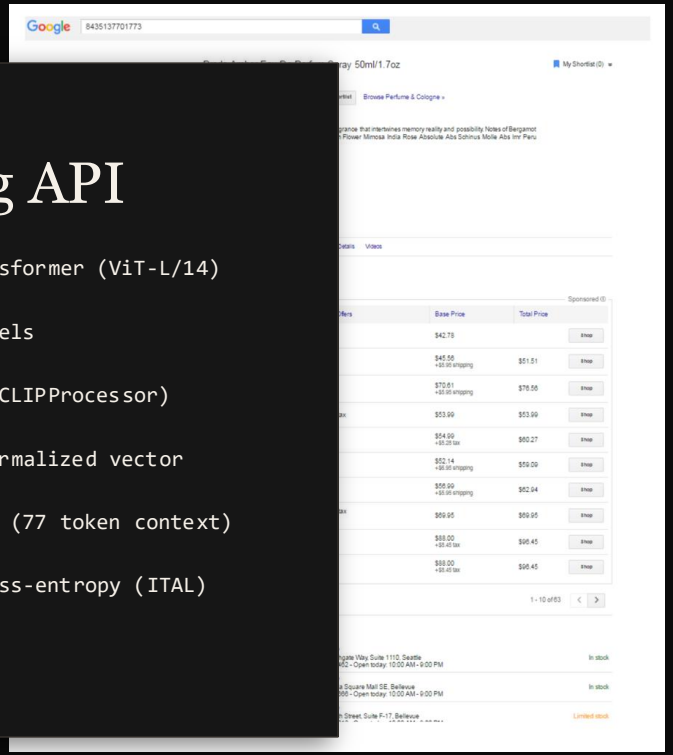
Google Shopping API

- Engine: Google Shopping via SerpAPI
- Query: '{name} perfume fragrance or eau de parfum
- Num results: per fragrance 20 product links
- Budget cap applied post-fetch

WEB SCRAPE

Google Shopping API

Engine:	Vision Transformer (ViT-L/14)
Query:	14 × 14 pixels
Num results:	224 × 224 (CLIPProcessor)
Budget:	768-d L2-normalized vector
	Transformer (77 token context)
	Softmax cross-entropy (ITAL)



Output — Descriptive Recommendations

TOP-3 RESULTS

SELECTION NO. 01

Aventus

Creed

Pineapple | Bergamot | Birch | Musk | 94%

"A daytime winter aesthetic — birch smoke projects success."

$320.00 Sephora · WITHIN BUDGET

SELECTION NO. 02

Oud Wood

Tom Ford

Oud | Sandalwood | Rosewood | Amber | 87%

"Formal evening silhouette — resinous warmth for smart casual."

$195.00 Nordstrom · WITHIN BUDGET

SELECTION NO. 03

Baccarat Rouge 540

Maison F. Kurkdjian

Jasmine | Saffron | Amberwood | Fir | 81%

"Luminous crystalline amber — commands every room."

$340.00 Bloomingdale's · WITHIN BUDGET

Summary & Future Directions

WHAT WE BUILT

Pipeline Summary

- ✓ LLM enriched dataset of 36,000 fragrances pulled across 5 datasets
- ✓ Zero-shot CLIP classifier with prompt-engineered vibe embeddings
- ✓ Sentence Transformer for semantic description matching
- ✓ 5-dimensional vibe representation of user inputs and fragrance data
- ✓ Recommendations using cosine similarity and optimized weights
- ✓ Real-time price discovery via SerpAPI Google Shopping engine
- ✓ Top 3 matching fragrances to match the user's outfit and occasion

FUTURE DIRECTIONS

Model Evolution

Dataset	Filter fragrance database for popular fragrances
Enrichment	Analyze fragrances individually for high accuracy
CLIP	Fine-tune CLIP on outfit data and associated vibes
all-MiniLM	Switch to Qwen3-Embedding for high dimensional embedding
Vibe	Increase vibe dimensions to represent fragrances better
Recommend	Switch to Qwen3-reranker for accuracy
Webscrape	Integrate Amazon API

Dress for the occasion. Scent it perfectly.